

PREDICTING OBESITY FROM LIFESTYLE DATA



Machine Learning & Data Mining | 2025/2026

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Group G | University of Cantabria | 21 May 2026



2,111

Records

Colombia · Peru · Mexico

16

Attributes

Demographic, dietary & lifestyle

NObeyesdad

Target

7 ordered obesity classes

PRE-PROCESSING PIPELINE

1

Encode

8 categorical → numeric (KNIME Category to Number)

2

Exclude

Height & Weight removed → prevent data leakage (BMI)

3

Normalise

Z-score on Age, FCVC, NCP, CH2O, FAF, TUE

4

Split

Stratified 80/20 train/test split

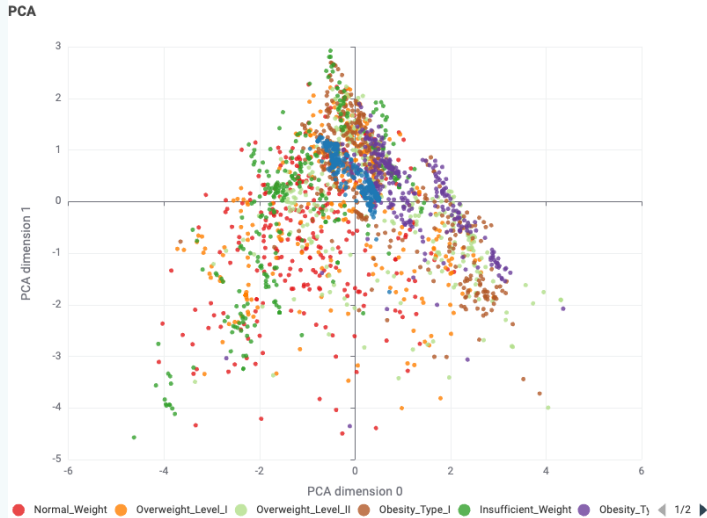
5

Note

77% synthetic data (SMOTE) — key limitation

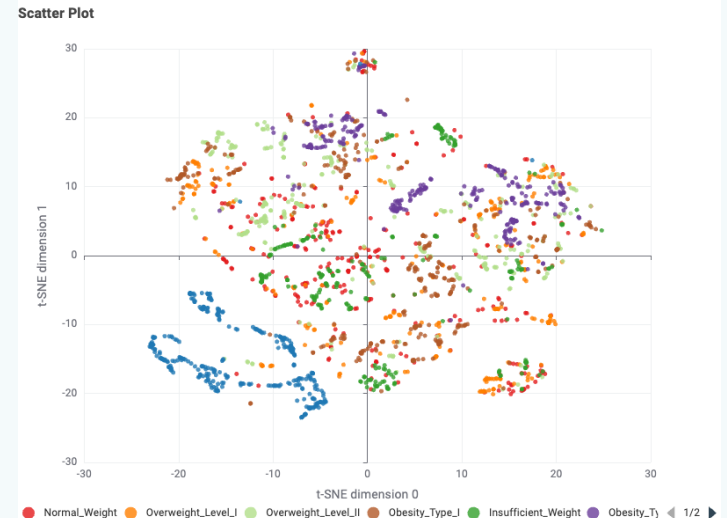
CLASS SEPARABILITY ANALYSIS

Figure 1: Dimensionality reduction projections coloured by obesity class.



(a) PCA (first two components)

- Heavy class overlap
- Linear boundaries are insufficient



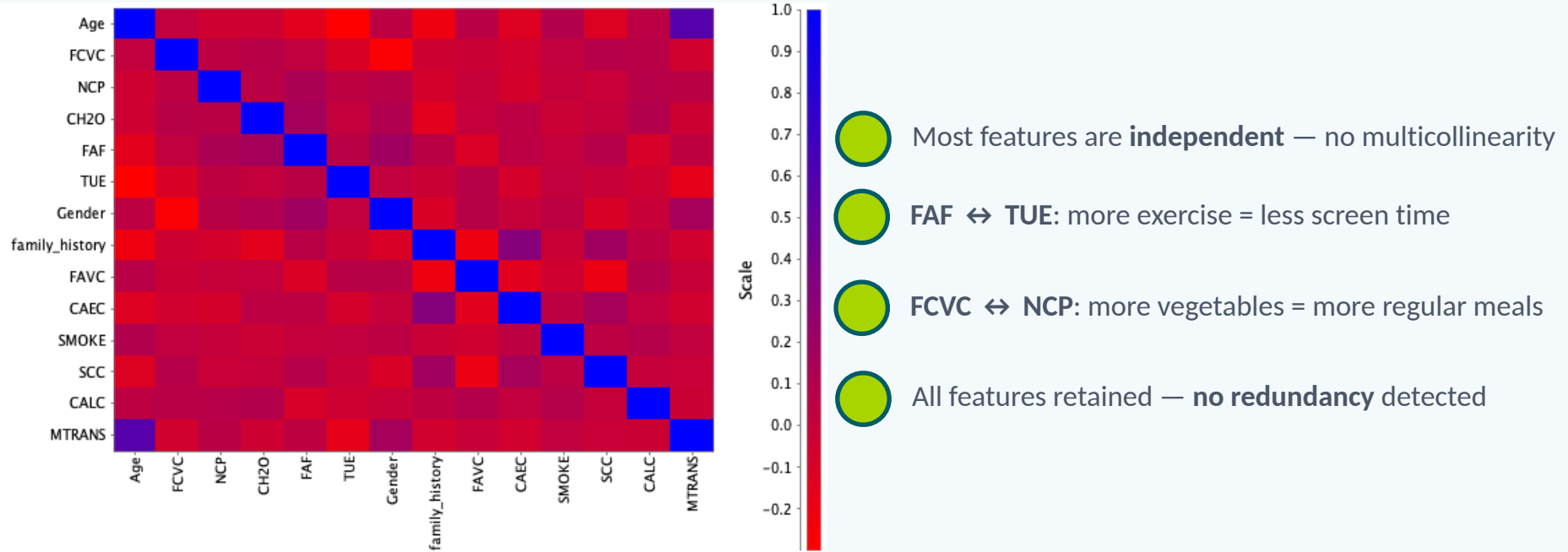
(b) t-SNE (perplexity 30).

- Clearer cluster structure visible for extreme classes
- Non-linear methods can separate the classes

★ Non-linear models are needed to distinguish all 7 obesity classes

FEATURE CORRELATION ANALYSIS

Figure 2: Covariance matrix of the continuous features.

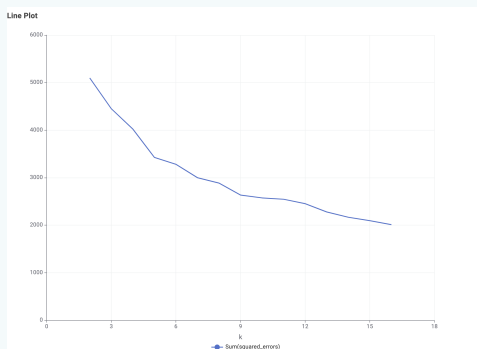
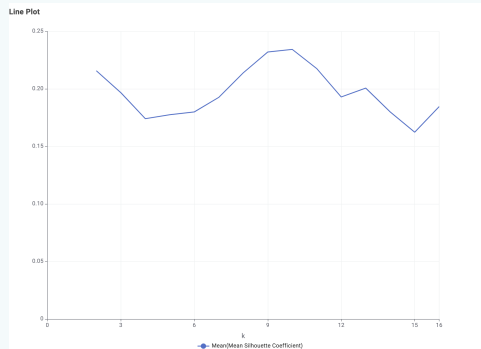


K-MEANS — CLUSTERING

k SELECTION

Elbow method + Silhouette coefficient

- Tested $k = 2$ to 16 ;
- The silhouette coefficient peaked at $k = 9, 10$;
- The elbow showed diminishing returns after $k = 9$;
- $k = 9$ was chosen.



KEY CLUSTERS

Cluster 7 **100%**

Obesity Type II

Cluster 4 **75.1%**

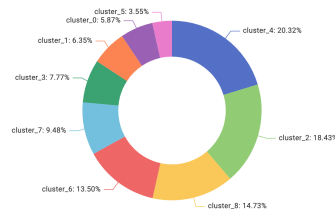
Obesity Type III · sedentary females

Cluster 3 **76.2%**

Insufficient Weight · young, low hydration

Cluster 1 **76.9%**

Normal Weight · active males

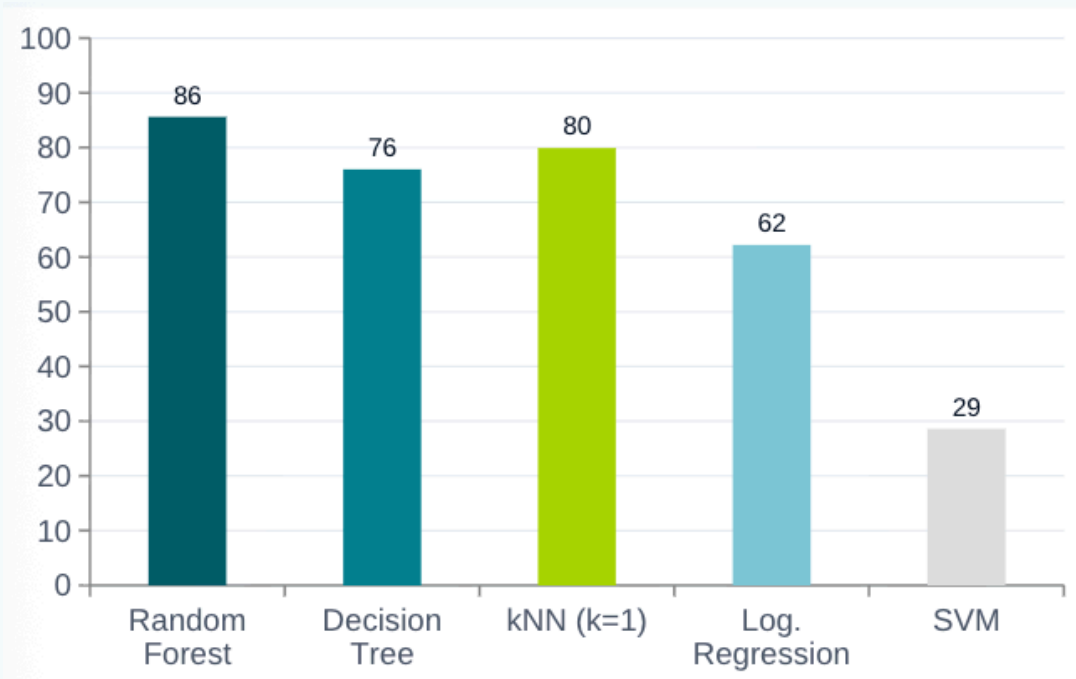


Distribution of clusters ($k = 9$).

✓ Lifestyle features contain enough structure to recover obesity categories — even without supervision.

CLASSIFICATION

Height & Weight excluded · Stratified 80/20 split · Baseline random = 14%



Random Forest

85.6%

Best model — ensemble captures non-linear patterns

Decision Tree

76.0%

Interpretable, solid performance

kNN k = 1

79.9%

SMOTE densifies neighbours → k=1 optimal

Logistic Reg.

62.2%

Reflects non-linear nature of the problem

SVM

28.6%

Poor — no kernel tuning, high dimensionality

NEURAL NETWORKS

RProp MLP (KNIME native)

Architecture:	3 hidden layers × 20 neurons
Inputs:	14 features (no Height/Weight)
Output:	7 obesity classes
Optimiser:	RProp (adaptive learning rate)
Accuracy:	72.3%

PER-CLASS RECALL

Obesity III		0.984
Insuff. Wt		0.790
Obesity I		0.681
Norm. Wt		0.600

H2O AutoML Deep Learning

Method:	Automated architecture search
Tuned:	Layers, neurons, dropout, L2
Accuracy:	75.7% (+3.4% vs RProp)
Cohen's κ:	0.715

IMPROVEMENTS OVER RProp

Insuff. Wt recall 0.790 → 0.975	+18.5 pp
Overweight I recall 0.623 → 0.935	+31.2 pp
Overall accuracy 72.3% → 75.7%	+3.4 pp

Rule 1

Adult Female + Public Transport → **Obesity Type III**

Lift 5.38

Confidence
82.6%

Rule 2

Adult + Sedentary ($\text{FAF} \leq 0.5$) + Public Transport → **Obesity Type III**

Lift 5.33

Confidence
81.8%

Rule 3

Sedentary + High Vegetable Intake → **Obesity Type III**

Lift 5.04

Confidence
77.3%



Correlation, not causation — no salad compensates a sedentary life

⊗ Sedentariness is the dominant factor across all rules — appearing in 5 of the top 6.

RESULTS & CONCLUSIONS

85.6%

Random Forest

79.9%

kNN k=1

75.7%

H2O AutoML

76.0%

Decision Tree

72.3%

RProp MLP

62.2%

Log. Regression

KEY FINDINGS

- Lifestyle data alone predicts obesity class — no weight or height needed
- Clustering without labels still recovers true obesity categories
- Sedentariness is the single strongest predictor for severe obesity
- Nudge Theory: design environments that make movement the default choice

Limitations: 77% synthetic data (SMOTE) · SVM not tuned · Association rules = correlation, not causality



Stockholm

Stockholm's piano staircase turned a subway exit into a musical instrument.

66% more people chose stairs over the escalator that day

THANK YOU

for your attention!



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